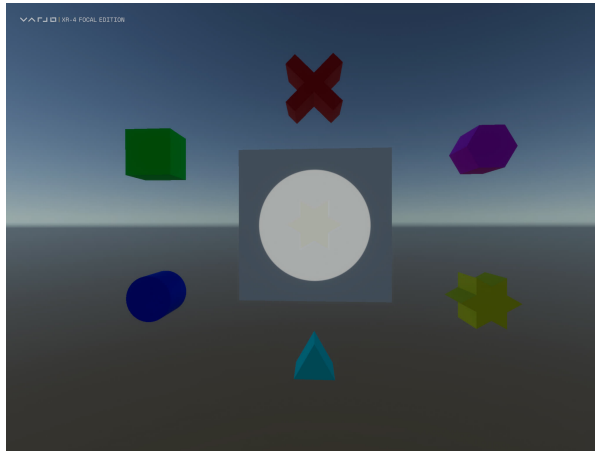


A Gaze Guidance Approach for Specialized Skill Training and Its Effect on Trainees Gaze Patterns

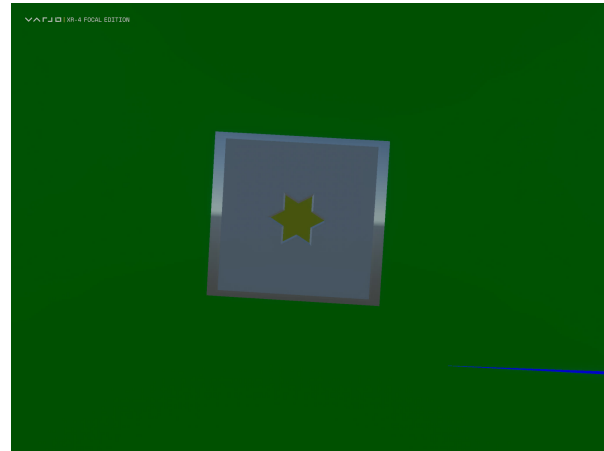
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(a) Flicker condition



(b) Mask condition

Figure 1: Presented gaze guidance

ABSTRACT

Efficient acquisition of gaze patterns is indispensable for improving specialized skills yet learning such patterns typically requires extensive time, repeated practice, and expert support. This paper proposes a gaze guidance approach for specialized skill training based on two visual designs, Flicker and Mask, which present guidance during practice to promote more efficient gaze behavior, and evaluates their effects on trainees gaze patterns. We evaluate the effects of these approaches on trainees gaze patterns in a virtual reality training environment. The results suggest that training time gaze guidance using Flicker and Mask can locally stabilize gaze behavior, but its benefits may be limited only in the training time and may not persist without continued support.

Index Terms: Gaze Guidance, Skill Acquisition, VR Training, Skill Training

1 INTRODUCTION

In improving specialized skills, efficient acquisition of gaze patterns is indispensable [1, 4, 6]. By directing attention to salient information at critical moments, performance and accuracy improve, and the failure rate decreases. These benefits from acquiring gaze patterns have been confirmed in various fields such as sports and medicine; for example, in one-handed square knot tying by surgical residents, it has been shown that acquiring an appropriate gaze pattern can suppress performance degradation even under high-anxiety conditions [1, 5]. However, acquiring efficient gaze patterns is more difficult than acquiring motor skills. Motor performance naturally improves in terms of speed and movement accuracy through repeated training. In contrast, gaze patterns require learning from

experience where to direct attention, and practicing this requires more time. Beginners are easily drawn to the instruments they are currently handling and the objects they are manipulating; as a result, their gaze tends to fixate on their hands, and anticipatory fixations are likely to be neglected, making improvement require conscious refinement of gaze and attention[3].

In supporting skill acquisition in medicine, training of gaze patterns has attracted attention. Law et al.[2] showed, using a VR task, that gaze distribution differs between experts and novices, and that experts look at the target more than the instruments. It was discussed that this is because experts are more accustomed to handling the instruments and therefore require less visual feedback of the instrument position than novices. In this study, in order to acquire instrument manipulation intervals that do not rely on visual feedback, we displayed gaze guidance to encourage keeping the gaze at the center even during instrument manipulation.

Vine et al.[7] showed that using highlight displays during Ball Pick-and-Drop, which is one of the one-handed tasks in laparoscopic surgical skills training, increases the amount of fixation on the ball and reduces the time required for completion. They also showed that this training leads to significantly improved performance, in both the amount of fixation and the time required for completion, in delayed assessment of the bimanual Grasp-and-Cut Task set as a transfer task. In this study, with application to microscope suturing training in mind, the guided position was set so that it would always be the center.

Yamaguchi et al.[8] proposed the Mask method, which hides the scene except for the central region when the trainee's gaze deviates during training, to keep gaze centered in virtual reality training and to facilitate the acquisition of instrument body ownership. However, in the comparison of performance before and after training, there were still no differences between conditions, and only one proposed method was examined. Therefore, in this study, we propose a Flicker method, which is displayed when the gaze deviates from the center of the task. While the Mask method is a method that

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imposes a penalty on the trainee when the gaze deviates, Flicker is a method that encourages attention to the center. By guiding the trainee's gaze back to the center in this way, the trainee acquires instrument manipulation ability that does not rely on visual feedback, and this promotes an increase in the amount of fixation on the central work area. We conducted a follow up experiment that introduced Flicker as an additional condition and examined its effects on gaze patterns and task performance.

2 IMPLEMENTATION

2.1 Hardware setup

This experiment was conducted using the XR headset Varjo XR-4. The gaze position was measured using the eye-tracking sensor embedded in the headset. This eye-tracking sensor outputs screen coordinates in the range from -1 to 1 with the center set to 0.

The virtual environment for the task was built using Unity. Participants manipulated the cube and the pieces using the Varjo left controller and right controller, respectively.

2.2 Implementation of the Proposed Method

When the trainee's gaze deviates from the position fixated by the expert during training, we correct the trainee's gaze pattern by displaying gaze guidance that returns the gaze position to its original location. This encourages the trainee to observe the center of the work area and to perform instrument manipulation in other areas using peripheral vision.

Specifically, we propose Flicker and a Mask as gaze guidance. Gaze guidance is displayed when the trainee's gaze deviates from the position that should be fixated. In the Flicker condition, a point flickering between white and black at 30 Hz is displayed at the position that should be fixated. In the Mask condition, a Mask that covers everything except the position that should be fixated is displayed.

3 EXPERIMENT

3.1 Purpose

To verify the effect of this system, which displays gaze guidance when the gaze deviates from the area that should be viewed, on improving trainees' gaze patterns, we conducted a user experiment.

3.2 Participants

A total of 12 participants (11 males and 1 female) took part in this experiment. The sample size was set in consideration of the number of conditions so that order effects could be counterbalanced. The participants' mean age was 23.6 years, and the variance was 3.14. Of the 12 participants, 11 had no experience using VR on a daily basis, and the remaining one had previously used VR on a daily basis but currently did so only about once every six months. Participants received 1,000 JPY as compensation after completing the study.

3.3 Task

In this experiment, we used the VR game described below as the task (Fig. 2).

At the center of the participant's field of view, a cube with holes of different colors and shapes on each face is placed, and around it, pieces with colors and shapes corresponding to each hole are displayed. Participants clear the game by fitting the correct piece into each of the six holes.

The operation is performed using controllers in both hands. By aiming the blue laser extending from the left-hand controller at the cube and tilting it up, down, left, and right while pressing a button, the cube can be rotated in any direction. By aiming the red laser extending from the right-hand controller at the target piece and tilting it up, down, left, and right or moving it forward and backward

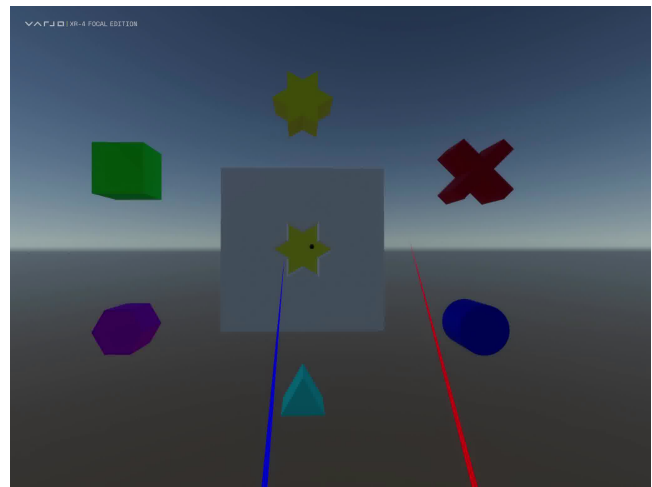


Figure 2: Game screen

while pressing a button, the piece can be moved in the corresponding direction. By facing the hole of the cube toward the front and bringing the piece closer, it becomes possible to fit it into the hole.

3.4 Conditions

In this experiment, we conducted the experiment under three conditions: Flicker, None, and Mask. In the Flicker condition, when the participant's gaze position is more than 18° away from the center in terms of visual angle, a blinking stimulus consisting of a white circular dot is displayed at the center, with a diameter equal to the length of one side of the cube as seen by the participant. This blinking periodically changes the alpha value between 0 and 1 at 30 Hz between white and black (Fig. 1a). The value of 18° is set so that the stimulus is displayed when the participant's gaze deviates from the cube. In the Mask condition, as in the Flicker condition, when the participant's gaze position is more than 18° away from the center in terms of visual angle, a Mask is displayed so that nothing except the cube can be seen (Fig. 1b). In the None condition, nothing is displayed even if the gaze deviates from the center. The experiment was conducted under each condition while considering order effects.

3.5 Procedure

This experiment was conducted using a within-subjects design.

First, the operation method was explained to the participants verbally. After that, using a trial version, they practiced manipulating the cube and the pieces, as well as fitting the pieces into the holes. The practice time was set to about 5 minutes, and during this time, participants were instructed to fixate on the cube during the task and to check the pieces using peripheral vision.

After the practice, in each condition, a pre-test, five training sessions, a post-test, and a questionnaire were conducted in this order. During task performance, participants were fixed using a chin rest to keep the orientation and height of the head constant.

3.6 Measurements

During training and testing, the following were recorded.

- Task completion time
- Gaze movements

In the post-experiment questionnaire, in addition to the System Usability Scale (SUS), participants were asked about the following three items Q1, Q2, and Q3, and responded on a 5-point scale from 1 to 5.

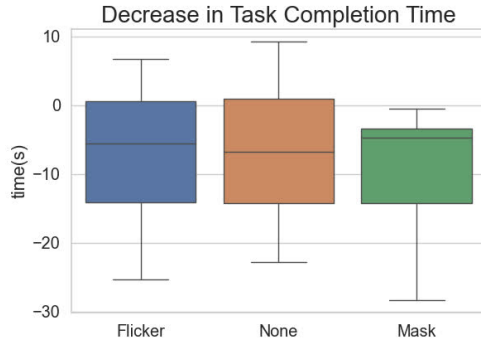


Figure 3: Decrease in task completion time

- Q1: Were you able to be aware of looking at the center due to this method?
- Q2: Did you feel that this method interfered with your concentration?
- Q3: Did you feel eye fatigue due to this method?

3.7 Hypotheses

In this experiment, we set the following hypotheses.

- H1: Even if gaze guidance is displayed when the gaze during training deviates from the center, the time required to complete the task does not change.
- H2: By displaying gaze guidance when the gaze during training deviates from the center, gaze dispersion during the task becomes smaller.

3.8 Results

3.8.1 Task Completion Time

Task completion time refers to the time from when the participant starts the game and the pieces are displayed around the cube until the participant finishes fitting the last piece into the cube.

When we applied the Wilcoxon signed-rank test to the task completion times in the pre-test and post-test for each condition, we obtained $p = 0.0455 (< 0.05)$ for Flicker, $p = 0.0455 (< 0.05)$ for None, and $p = 0.0254 (< 0.05)$ for Mask, indicating that task completion time was significantly shortened in all conditions.

The mean decrease in task completion time from the pre-test to the post-test in each condition was 6.933 seconds for Flicker, 7.677 seconds for None, and 8.364 seconds for Mask (Fig. 3). Using the Friedman test, we obtained $p = 0.92 (> 0.05)$, and no significant difference was observed.

The mean task completion time during training in each condition was 34.972 seconds for Flicker, 32.691 seconds for None, and 33.85 seconds for Mask. Using the Friedman test, we obtained $p = 0.638 (> 0.05)$, and no significant difference was observed.

3.8.2 Standard Deviation of Gaze Position

In each condition, the mean decreases in the standard deviation of gaze position in the x- and y-axis directions from the pre-test to the post-test were 0.0019 and 0.0026 for Flicker, -0.0098 and -0.011 for None, and 0.012 and 0.011 for Mask (Fig. 4). Using the Friedman test, we obtained $p = 0.263 (> 0.05)$ for the x-axis direction and $p = 0.174 (> 0.05)$ for the y-axis direction, and no significant differences were observed. In addition, when we extracted the 10% with large movements of the right controller (piece manipulation), where the gaze tends to move greatly, and applied the Friedman test, we obtained $p = 0.368 (> 0.05)$ for the x-axis direction and

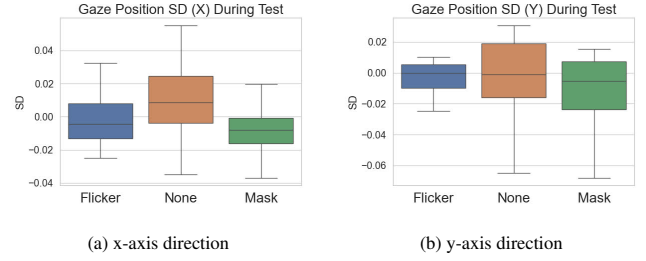


Figure 4: Decrease in gaze standard deviation during testing

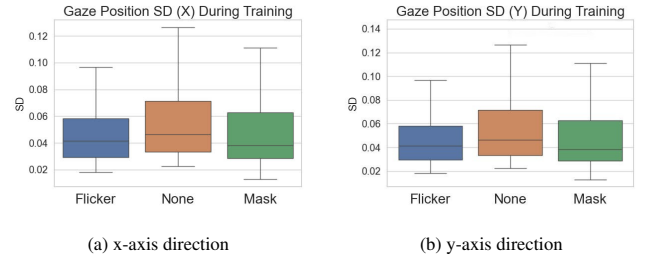


Figure 5: Gaze standard deviation during training

$p = 0.717 (> 0.05)$ for the y-axis direction, and no significant differences were observed.

The mean standard deviations of gaze position in the x- and y-axis directions during training in each condition were 0.048 and 0.041 for Flicker, 0.055 and 0.046 for None, and 0.048 and 0.035 for Mask (Fig. 5). Using the Friedman test, we obtained $p = 0.54 (> 0.05)$ for the x-axis direction and $p = 0.0269 (< 0.05)$ for the y-axis direction, indicating that a significant difference was observed only in the y-axis direction. The Wilcoxon signed-rank test results were $p = 0.11$ between Flicker and None, $p = 0.22$ between Flicker and Mask, and $p = 0.081$ between None and Mask, and no significant differences were observed in the post hoc tests. In addition, when we extracted the 10% with large right-controller movements during training and applied the Friedman test, we obtained $p = 0.861 (> 0.05)$ for the x-axis direction and $p = 0.705 (> 0.05)$ for the y-axis direction, and no significant differences were observed.

3.8.3 Error Distance

Error distance refers to the distance from the region where gaze guidance is not displayed when the gaze deviates from the range that should be viewed.

In each condition, the mean decrease in error distance from the pre-test to the post-test was -0.0107 for Flicker, 0.0091 for None, and 0.00196 for Mask (Fig. 6). Using the Friedman test, we obtained $p = 0.7704 (> 0.05)$, and no significant difference was observed.

The mean error distance during training in each condition was 0.06177 for Flicker, 0.075 for None, and 0.0356 for Mask. Using the Friedman test, we obtained $p = 0.0096 (< 0.05)$, indicating a significant difference. In the post hoc tests using the Wilcoxon signed-rank test, significant differences were observed between the Flicker and None conditions ($p = 0.0242$) and between the None and Mask conditions ($p = 0.00061$), while no significant difference was observed between the Flicker and Mask conditions ($p = 0.19$).

3.8.4 Questionnaire

The SUS scores in each condition were 78.96 for Flicker, 76.46 for None, and 70.42 for Mask. Using the Friedman test, we obtained $p = 0.541 (> 0.05)$, and no significant difference was observed.

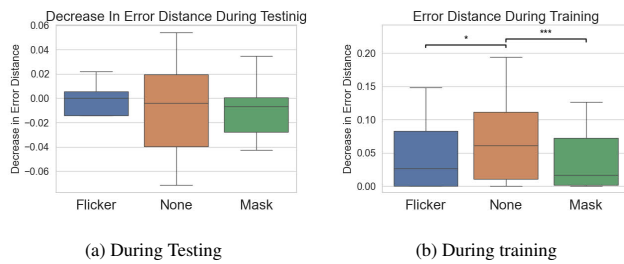


Figure 6: Error distance

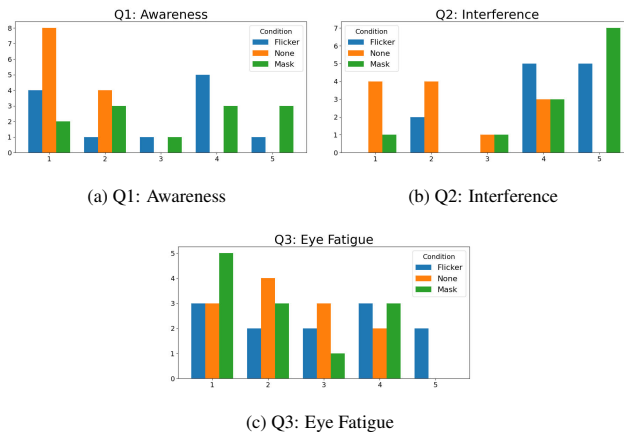


Figure 7: Results of the additional questionnaire

served among the three conditions. In addition, no significant differences were observed among the conditions for each question.

When we applied the Friedman test to the responses to the additional questionnaire, significant differences were observed among the conditions for all of Q1, Q2, and Q3, with $p = 0.0188 (< 0.05)$, $p = 0.00343 (< 0.05)$, and $p = 0.00403 (< 0.05)$, respectively. When we conducted post hoc tests using the Wilcoxon signed-rank test for each question, significant differences were observed between the None and Flicker conditions and between the None and Mask conditions for Q1 and Q2, while no significant difference was observed between the Flicker and Mask conditions. For Q3, significant differences were observed between the Flicker and None conditions and between the Flicker and Mask conditions, while no significant difference was observed between the None and Mask conditions (Fig. 7).

4 DISCUSSION

The proposed method did not have a significant effect on task completion time. This is consistent with the result that motor performance matures faster than the acquisition of efficient gaze patterns, and the results of this experiment supported H1. In contrast, gaze guidance during training did not affect gaze dispersion during the task, and H2 was not supported. In addition, no differences were observed among the conditions in terms of error distance. This is considered to be because, even if gaze was guided, it did not guide participants to the visual information at the guided position and thus could not influence gaze patterns.

Gaze guidance during training reduced gaze dispersion in the y-axis direction and reduced the distance when the gaze deviated from the range that should be viewed. This is considered to be because the display of gaze guidance when the gaze position deviated served as an indication to the user. However, when the movement of the right controller, which tends to cause large gaze movements, was

large, it is considered that even the indication provided by displaying gaze guidance did not affect the user's gaze patterns.

According to the questionnaire results, participants responded that they were able to be more aware of looking at the center in the Flicker and Mask conditions compared to the None condition. However, participants also responded that the Flicker and Mask conditions interfered with concentration and caused eye fatigue compared to the None condition.

Limitations. Our participant sample was imbalanced in gender (11 males and 1 female) and exhibited a narrow age range ($M = 23.6$ years; variance = 3.14, $SD \approx 1.77$). These characteristics may limit the generalizability of the findings, as gaze strategies and susceptibility to visual guidance can vary across demographic groups. Future studies should replicate the experiment with a more gender-balanced and age-diverse cohort.

Summary and Future Work. In summary, Flicker and Mask did not lead to condition-level improvements in task completion time, gaze dispersion during testing, or error distance during testing, whereas both methods showed benefits during training by locally stabilizing gaze behavior (reduced y-axis dispersion) and by reducing gaze deviations outside the intended viewing range. Participants also reported higher awareness of keeping gaze centered, but at the cost of increased interference with concentration and eye fatigue. A key direction for future work is to design less intrusive and more adaptive guidance—e.g., modulating guidance strength based on hand/controller movement—and to evaluate longer-term retention and transfer effects after extended practice.

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